Tutorial 1 – Scalability and Synchronization

Deadline: 17.05.2026

Objectives:

- Get acquainted with scaling of response times
- Investigate a method to synchronize an asynchronous swarm
- Experience influence of swarm connectivity/swarm density

1 Scaling of a Computer System

We create a simple model of a computer system. The following is a simple example from queuing theory. We assume that new jobs for the computer system come in with a constant rate of α and are added to a waiting list. We also assume that the process is without memory, that is, incoming jobs are statistically independent from each other. Therefore, we model the incoming jobs as a Poisson process. The probability that i jobs arrive within a given time interval Δt is given by:

$$P(X = i) = \frac{e^{-\lambda}\lambda^i}{i!}$$

for $\lambda = \alpha \Delta t$. In the following we simply set $\Delta t = 1$.

a) Plot $P(X = i)$ for a reasonable interval of X and $\alpha \in \{0.01, 0.1, 0.5, 1\}$.

b) Implement a program that samples numbers of incoming jobs from $P(X = i)$.

c) Implement a model that iterates over these two phases: calculate and manage new incoming jobs, then operate on the current job (one at a time). Generate a sample of 2000 time steps for $\alpha = 0.1$ and a processing duration of 4 steps per job. What is the average length of the waiting list?

d) Modify your program so that you can repeat the 2000-step simulation many times and average the resulting queue lengths. For each rate $\alpha \in [0.005, 0.25]$ in steps of 0.005, run the model 200 independent times, compute the average queue length for each run, then average these 200 values to obtain one final number per α. Plot this final average as a function of α. Discuss what this reveals about the system's behavior.

e) Do the same for a processing duration of only 2 steps per job with rates $\alpha \in [0.005, 0.5]$ in steps of 0.005. Compare the two plots and discuss the change in the system's behavior as a result of these changes.

2 Synchronization of a Swarm

Swarm systems are asynchronous systems. There is no central clock that could be accessed by everyone. If a swarm needs to act in synchrony, it has to explicitly synchronize first. An example of a biological system that shows synchronization is a population of fireflies (see Fig. 1).

Figure 1: Synchronized flash of fireflies

In the following we create a simple model of such a firefly population. The population is scattered randomly over a $1{\times}1$ square (uniform distribution). We assume that fireflies are stationary and can only perceive local neighbors in their vicinity. We say two fireflies are within the vicinity of each other if the distance between them is smaller than r. Hence, there is a virtual disc centered at each firefly's position and every other firefly located within that disc is considered a neighbor. The fireflies flash in cycles. We define the cycle length by $L = 50$ time steps. We initialize the fireflies to uniformly randomly distributed clock cycles (i.e., not synchronized). The firefly flashes for $L/2$ steps followed by $L/2$ steps of not flashing. This holds except when the firefly attempts to correct its cycle to synchronize. In the time step after it has started to flash it checks its neighbors and tests whether the majority of them is actually already flashing. If so, the firefly corrects its clock by adding 1, that is, it is decreasing the current flashing cycle from $L/2$ to $L/2 - 1$ steps and will consequently flash 1 step earlier next time.

a) Implement the model for swarm size $N = 150$ and cycle length $L = 50$. Calculate the average number of neighbors per firefly for vicinity distances $r \in \{0.05, 0.1, 0.5, 1.4\}$. Plot the number of currently flashing fireflies over time for vicinity distances $r \in \{0.05, 0.1, 0.5, 1.4\}$ for 5000 time steps each. When plotting the number of currently flashing flies, make sure you plot the full interval of $[0, 150]$ for the vertical axis.

b) Extend your model to determine the minimum and maximum number of concurrently flashing fireflies during the very last cycle (last $L = 50$ time steps starting from $t = 4950$). By subtracting the minimum from the maximum you get twice the amplitude of the flash cycle. Average the measured amplitudes over 50 samples each (50 independent simulation runs with 5000 time steps each) and plot them over vicinities $r \in [0.025, 1.4]$ in steps of 0.025. What seems to be a good choice for the vicinity and the swarm density? What do low amplitudes tell us about synchronization, and what do high amplitudes reveal?

Submission Requirements:

- Groups can consist of a maximum of 2 students
- Submit one ZIP file named:
  `CollRob_01_YOURLASTNAME1_YOURLASTNAME2.zip`
- Include only necessary files (no large or irrelevant files)
- Ensure the project is **well-structured and clean**
- Include all source code, runnable without complex setup
- Provide a short README, explaining how to run the code
- Include plots for each subtask (PNG or JPG)
- Include a report (including implementation, results, group members, and completed tasks)
- Include a video explanation of your code (.mp4, required)
- Do not include the original task sheet in the submission